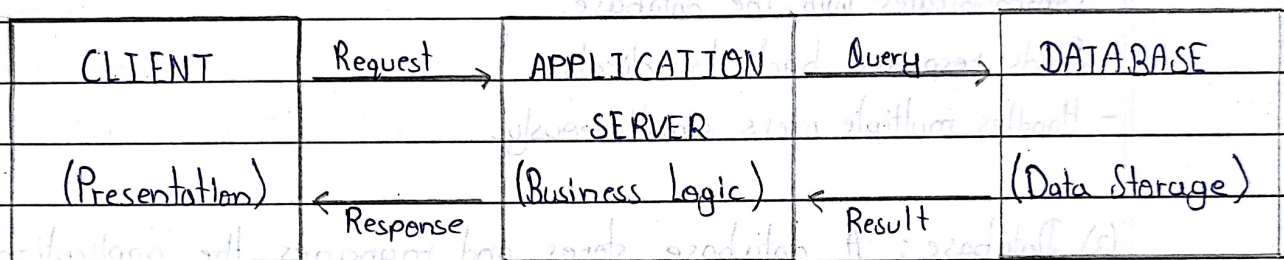


26/07/2026

Assignment - 1

1) Explain Client-Server Architecture in detail with a neat diagram. Describe the role of Client, Server and Database.

→ Client-Server Architecture is a network model in which multiple clients request services or resources from a centralized server. The server processes the requests, accesses the database if necessary, and sends the required response back to the client. It is the Foundation of modern web applications such as banking systems, e-commerce websites, social media platforms, and online learning portals.



Laptop/Mobile

Web/API Server

MySQL/PostgreSQL

Diagram: Client - Server Architecture

→ Components of Client-Server Architecture:

(1) Client: A client is a computer, mobile device, or application that sends requests to the server to access information or services.

Examples: Google Chrome, Mozilla Firefox, Microsoft Edge, Mobile Applications.

Functions of Client:

- Provides user Interface.
- Accepts user input.
- Sends requests to the server.

- Receives and displays response.
- Executes client-side scripts like JavaScript.

(2) Server: A server is a powerful computer or software that processes client requests and provides the requested resources.

Examples: Apache Server, Nginx, Microsoft IIS, Node.js Server.

Functions of Server:

- Accepts client requests.
- Processes business logic.
- Authenticate users.
- Communicates with the database.
- Sends response back to clients.
- Handles multiple users simultaneously.

(3) Database: A database stores and manages the application's data in an organized manner.

Examples: MySQL, PostgreSQL, Oracle, SQL Server, MongoDB.

Functions of Database:

- Stores user information.
- Stores transaction records.
- Retrieves requested data.
- Updates existing records.
- Deletes unwanted data.
- Maintains data integrity and security.

→

Client-Server Architecture provides efficient communication between clients and servers. It ensures centralized management, better security, scalability, and reliable access to shared resources, making it suitable for almost all modern web applications.

2) Differentiate Static and Dynamic Websites. Explain their advantages, disadvantages and applications.

→ A website is a collection of interconnected web pages accessible through the internet. Websites are mainly classified into:

- Static Website
- Dynamic Website

(1) Static Website: A static website displays fixed content. Every user sees the same webpage unless the developer manually updates the HTML files.

Characteristics:

- Fixed content.
- Developed using HTML and CSS.
- No database required.
- Faster loading.
- Easy to host.

(2) Dynamic Website: A dynamic website generates content according to user requests using server-side programming languages and databases.

Characteristics:

- Content changes dynamically.
- Uses database.
- Server-side scripting.
- Interactive
- Supports login systems.

→ Advantages of Static Website:

- Simple to create.
- Fast loading speed.

- Better security.
- Low development cost.
- Easy deployment.
- Less server resources.
- Suitable for beginners.

→ Disadvantages of Static Website:

- Manual updates required.
- No user interaction.
- Cannot manage large data.
- No login functionality.
- Difficult to scale.
- Limited functionality.

→ Applications of Static Website:

- Personal Portfolio
- Resume Website
- Company Profile
- College Notice Board
- Event Website
- Product Brochure
- Landing Pages

→ Advantages of Dynamic Website:

- Interactive user interface.
- Easy content management.
- Personalized content.
- Database support.
- Online transactions possible.
- User authentication.
- Easy scalability.

- Disadvantages of Dynamic Website:
- Complex development.
 - Higher hosting cost.
 - Security risks.
 - Requires database management.
 - More maintenance.
 - Slower than static websites.

- Applications of Dynamic Website:
- Facebook
 - Instagram
 - Amazon
 - Flipkart
 - Gmail
 - Hospital Management System

3) Discuss the role of Web Technologies in Banking and Finance. Explain how modern banking systems use web technologies.

↳ Role of Web Technologies in Banking and Finance: Web technologies have transformed the traditional banking system into a digital platform where customers can perform financial transactions anytime and from anywhere.

(1) Internet Banking: Customers can access their bank accounts using a web browser without visiting the bank.

Services include: Balance enquiry, Mini-statement, Fund transfer, Bill payments, Fixed deposit creation, Loan applications.

(2) Online Fund Transfer: Web applications allow secure transfer of money between bank accounts.

Examples: NEFT, RTGS, IMPS, UPI.

(3) Online Bill Payments: Customers can pay various bills through the banking websites.

Examples: Electricity bill, Water bill, Gas bill, Mobile Recharge, DTH Recharge, Broadband Bills.

(4) Digital Payments: Banks provide secure payment gateways for online shopping and merchant transactions.

Examples: Credit Card, Debit Card, Internet Banking, QR Code Payments, UPI Payments.

(5) Online Loan Services: Customers can apply for loans, upload documents, check loan eligibility, track application status, view EMI schedules.

(6) Investment Services: Banks provide investment facilities through the web applications.

Examples: Mutual Funds, FD's, Government Bonds, SIP Investments, Insurance Plans.

→ How Modern Banking Systems Use Web Technologies:

Step-1: User Login

The customer opens the banking website and enters login credentials.

Step-2: Authentication

The server verifies: Username, Password, OTP, Biometric Authentication (if available).

Step-3: Server Processing

The web server processes the customer's request.

Step-4: Database Access

The server retrieves or updates account information stored in the database.

Step-5: Response

The requested information is displayed securely on the user's browser.

→ Technologies Used in Banking:

Frontend: HTML, CSS, Javascript, React, Angular

Backend: Java, Python, PHP, ASP.NET, Node.js

Database: MySQL, Oracle, PostgreSQL, SQL Server.

Security Technologies: SSL/TLS, HTTPS, Encryption, Digital Signature, Firewall, OTP Authentication, Two-Factor Authentication (2FA).

4) What is CSS? Explain different types of CSS. List out advantages and the disadvantages of CSS.

→ CSS (Cascading Style Sheets) is a stylesheet language used to control the appearance and layout of HTML documents. It separates the presentation layer from the content, making web pages more attractive, consistent, and easier to maintain.

→ Types of CSS:

CSS can be applied to HTML documents in three different ways.

Inline CSS

Internal CSS

External CSS

style = " "

<style>

style.css

(1) Inline CSS : It is written directly inside the HTML element using the style attribute.

Syntax: `<p style="color:blue; Font-size: 20px;">`

Welcome to WAD

`</p>`

(2) Internal CSS : It is written inside the `<style>` tag within the `<head>` section of an HTML page.

Syntax: `<html>`

`<head>`

`<style>`

`h1{`

`color:red;`

`text-align:center;`

`}`

`</style>`

`</head>`

`<body>`

`<h1> Welcome </h1>`

`</body>`

`</html>`

(3) External CSS : It is written in a separate .css file and linked to HTML using the `<link>` tag.

HTML File: `<link rel="stylesheet" href="style.css">`

style.css: `body{`

`background-color:lightblue;`

`}`

`h1{`


```
color: darkblue;  
text-align: center;  
}
```

→ Advantages of CSS:

- Separates content from design.
- Reduces HTML code.
- Improves website appearance.
- Faster page loading due to reusable styles.
- Easy maintenance.
- Consistent design across pages.
- Supports responsive web design.
- Improves user experience.
- Easy to modify layouts.

→ Disadvantages of CSS:

- Website design
- Responsive web development
- Landing pages
- E-commerce websites
- Portfolio websites
- Educational websites
- Banking websites
- Government portals.

5) Create an HTML Form for student registration using textbox, checkbox, radio button, combo box, file upload, submit and reset button.

→ An HTML Form is used to collect information for users. It contains various input controls such as text boxes, radio buttons, checkboxes, drop-down lists, file upload controls, and buttons. The collected data can be submitted to a web

server for processing.

→ HTML Program:

Code:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title> Student Registration Form </title>
```

```
<style>
```

```
body {
```

```
font-family: Arial;
```

```
background: #F2F2F2;
```

```
}
```

```
form {
```

```
width: 500px;
```

```
margin: auto;
```

```
background: white;
```

```
padding: 20px;
```

```
border-radius: 10px;
```

```
box-shadow: 0px 0px 10px gray;
```

```
}
```

```
h2 {
```

```
text-align: center;
```

```
color: darkblue;
```

```
}
```



```
table {
```

```
    width: 100%;
```

```
}
```

```
td {
```

```
    padding: 8px;
```

```
}
```

```
input, select {
```

```
    width: 100%;
```

```
    padding: 6px;
```

```
}
```

```
input [type = radio],
```

```
input [type = checkbox] {
```

```
    width: auto;
```

```
}
```

```
input [type = submit],
```

```
input [type = reset] {
```

```
    width: 100px;
```

```
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<form>
```

```
<h2> Student Registration Form </h2>
```

```
<table>
```

```
<tr>
```

```
<td> Name </td>
```

```
<td> <input type = "text" placeholder = "Enter Name" > </td>
```

```
</tr>
```

```
<tr>
```

```
<td> Enrollment No. </td>
```

```
<td> <input type = "text" > </td>
```

```
</tr>
```

```
<tr>
```

```
<td> Email </td>
```

```
<td> <input type = "email" > </td>
```

```
</tr>
```

```
<tr>
```

```
<td> Password </td>
```

```
<td> <input type = "password" > </td>
```

```
</tr>
```

```
<tr>
```

```
<td> Gender </td>
```

```
<td>
```

```
<input type = "radio" name = "gender" > Male
```

```
<input type = "radio" name = "gender" > Female
```

```
</td>
```

```
</tr>
```


<tr>

<td> Hobbies </td>

<td>

<input type = "checkbox"> Reading

<input type = "checkbox"> Sports

<input type = "checkbox"> Music

</td>

</tr>

<tr>

<td> Department </td>

<td>

<select>

<option> Select Department </option>

<option> Computer Engineering </option>

<option> Information Technology </option>

<option> Mechanical Engineering </option>

<option> Civil Engineering </option>

<option> Electrical Engineering </option>

</select>

</td>

</tr>

```
<tr>
```

```
<td> Upload Photo </td>
```

```
<td>
```

```
<input type = "file">
```

```
</td>
```

```
</tr>
```

```
<tr>
```

```
<td></td>
```

```
<td>
```

```
<input type = "submit" value = "Submit">
```

```
<input type = "reset" value = "Reset">
```

```
</td>
```

```
</tr>
```

```
</table>
```

```
</form>
```

```
</body>
```

```
</html>
```


Output:

Student Registration Form

Name Enrollment No. Email Password Gender ☐ Male ☐ FemaleHobbies ☐ Reading☐ Sports☐ MusicDepartment Upload Photo

→ HTML Forms provide an efficient way to collect user information. By using various form controls, developers can create interactive and user-friendly web applications.

6) What is HTML table? Explain table element with necessary attributes and write code to produce a given table layout.

→ An HTML Table is used to arrange data in rows and columns. Tables are created using the `<table>` element along with tags such as `<tr>`, `<th>` and `<td>`. HTML tables are commonly used to display structured information like marksheets, timetables, employee records, and reports.

Basic Structure of an HTML Table

`<table>`

`<tr>`

`<th> Heading </th>`

`</tr>`

`<tr>`

`<td> Data </td>`

`</tr>`

`</table>`

→ HTML Table Elements:

Element	Description
<code><table></code>	Creates a table.
<code><tr></code>	Defines a table row.
<code><th></code>	Defines a table heading.
<code><td></code>	Defines table data.
<code><caption></code>	Adds table title.
<code><thead></code>	Groups table header.
<code><tbody></code>	Groups table body.
<code><tfoot></code>	Groups table footer.

→ Important Table Attributes:

Attribute	Purpose
border	Displays border around table.
cellpadding	Space between cell border and content.
cellspacing	Space between adjacent cells.
rowspan	Merges rows.
colspan	Merges columns.
width	Sets table width.
height	Sets table height.
align	Aligns table.
bgcolor	Sets background color.

→ HTML Program:

Code:

`<!DOCTYPE html>``<html>``<head>``<title> Student Table </title>``<style>``table {``border-collapse: collapse;``width: 60%;`

```
margin: auto;  
}
```

```
th, td {
```

```
border: 2px solid black;
```

```
padding: 10px;
```

```
text-align: center;
```

```
}
```

```
th {
```

```
background: lightblue;
```

```
}
```

```
caption {
```

```
font-size: 25px;
```

```
font-weight: bold;
```

```
margin-bottom: 10px;
```

```
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<table>
```

```
<caption> Student Details </caption>
```

```
<tr>
```

```
<th> Roll No </th>
```

```
<th> Name </th>
```

```
<th> Department </th>
```


<th> Semester </th>

</tr>

<tr>

<td> 101 </td>

<td> Rahul </td>

<td> Computer </td>

<td> 5 </td>

</tr>

<tr>

<td> 102 </td>

<td> Priya </td>

<td> IT </td>

<td> 5 </td>

</tr>

<tr>

<td> 103 </td>

<td> Amit </td>

<td> Mechanical </td>

<td> 5 </td>

</tr>

</table>

</body>

</html>

Output:

Student Details

Roll No	Name	Department	Semester
101	Rahul	Computer	5
102	Priya	IT	5
103	Armit	Mechanical	5

- 7) Write HTML and CSS code: (i) Set background color for active link to 'blue' (ii) Set unordered list style to 'circle', (iii) Set animation.gif as background image.

→ HTML is used to create the structure of a webpage, while CSS is used to control its appearance and layout. CSS provides different properties to style hyperlinks, lists, backgrounds, fonts, and other HTML elements.

- (i) Set Background Color for Active Link to Blue

Code:

<!DOCTYPE html>

<html>

<head>

<title> Active Link Example </title>

<style>

a:link {

color: black;


```
text-decoration: none;  
}
```

```
a: visited {  
  color: green;  
}
```

```
a: hover {  
  color: red;  
}
```

```
a: active {  
  background-color: blue;  
  color: white;  
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<h2> Active Link Example </h2>
```

```
<a href = "https://www.google.com">
```

```
Click Here
```

```
</a>
```

```
</body>
```

```
</html>
```

Output:

Active Link Example

Click Here

(ii) Set Unordered List Style to 'Circle'

Code:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
    <title> Circle List </title>
```

```
<style>
```

```
ul {
```

```
list-style-type: circle;
```

```
font-size: 20px;
```

```
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<h2> Programming languages </h2>
```



```
<ul>
```

```
<li> HTML </li>
```

```
<li> CSS </li>
```

```
<li> JavaScript </li>
```

```
<li> Python </li>
```

```
</ul>
```

```
</body>
```

```
</html>
```

Output:

Programming Languages

- HTML

- CSS

- JavaScript

- Python

(iii) Set animation.gif as Background Image

Code:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title> Background Image </title>
```

```
<style>
```

```
<body>
```

```
body {
```

```
background-image: url ("animation.gif");
```

```
background-repeat: no-repeat;
```

```
background-size: cover;
```

```
background-attachment: fixed;
```

```
}
```

```
h1 {
```

```
text-align: center;
```

```
color: white;
```

```
margin-top: 250px;
```

```
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<h1> Welcome to Web Application Development </h1>
```

```
</body>
```

```
</html>
```


- 8) Write CSS code to implement: (1) Ordered list with red border, padding, Font-size 20. (2) Use Flower.jpg as background. (3) Change background on text box focus. (4) Create fixed position link at bottom.

→ CSS (Cascading Style Sheets) is used to enhance the appearance of HTML elements by applying styles such as borders, colors, padding, positioning, backgrounds, and animations.

(1) Ordered List with Red Border, Padding and Font Size 20px

Code:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title> Ordered List </title>
```

```
<style>
```

```
ol {
```

```
border: 3px solid red;
```

```
padding: 20px;
```

```
font-size: 20px;
```

```
width: 250px;
```

```
}
```

```
</style>
```

```
</head>
```

```
<body>
```

<h2> Subjects </h2>

 HTML

 CSS

 JavaScript

 PHP

</body>

</html>

Output :

Subjects

1. HTML
2. CSS
3. JavaScript
4. PHP

(2) Use Flower.jpg as Background.

Code:

<!DOCTYPE html>

<html>

<head>

<title> Background </title>

<style>

body {

background-image: url ("flower.jpg");

background-repeat: no repeat;

background-size: cover;

background-attachment: fixed;

}

</style>

</head>

<body>

<h1> Flower Background Example </h1>

</body>

</html>

(3) Change Background Color on Textbox Focus.

Code:

<!DOCTYPE html>

<html>

<head>

<title> Textbox Focus </title>

<style>

```
input [type = text]: focus {
```

```
background-color : yellow;
```

```
color : black ;
```

```
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<h2> Enter Your Name </h2>
```

```
<input type = "text" placeholder = "Enter Name" >
```

```
</body>
```

```
</html>
```

Output:

Enter Your Name

Enter Name

(4) Create Fixed Position Link at the Bottom

Code:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title> Fixed Link </title>
```

```
<style>
```

```
.fixedlink {
```

```
position: fixed;
```

```
bottom: 10px;
```

```
right: 20px;
```

```
background: blue;
```

```
color: white;
```

```
padding: 10px 20px;
```

```
text-decoration: none;
```

```
border-radius: 5px;
```

```
}
```

```
</style>
```

</head>

<body>

<h2> Fixed Position Link Example </h2>

<p> Scroll the page and notice the link remains fixed at bottom. </p>

Visit Google

</body>

</html>

Output:

Fixed Position Link Example

Scroll the page and notice the link remains fixed at the bottom.

Visit Google